Vivek Gupta

Backend & AI Engineer · RAG & LLM Evaluation · Python · C# / .NET · Distributed Systems

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PROFESSIONAL SUMMARY

Backend Engineer with 4 years building distributed systems, event-driven microservices, and REST APIs in production as the sole backend engineer at PPI Advisory in Toronto, a national Canadian financial advisory firm, owning system design, testing, CI/CD, and production support for services used by 600+ employees. Now building LLM applications in Python: designed, evaluated, and deployed a RAG assistant over the FastAPI docs, raising answer correctness from 26% to 87% through ten measured experiments. Proficient in Python, C#, .NET, Node.js, TypeScript, SQL Server, MongoDB, and Azure.

KEY ACHIEVEMENTS

- **Built and deployed a RAG system with its own evaluation harness** — LLM-as-judge scoring raised answer correctness from 26% to 87% and faithfulness from 35% to 83%.
- **Sole engineer on a distributed email system** processing 60,000–80,000 events per day for 600 employees, with zero data loss even during API throttling.
- **Eliminated recurring licensing costs** by building an in-house replacement for a paid third-party plugin and automating lifecycle management across 100-150 accounts.
- **Owned a SQL Server commission engine end-to-end**, from initial architecture through production support, serving hundreds of financial advisors nationwide, with results surfaced through Power BI dashboards.

AI ENGINEERING PROJECT

RAG Docs Assistant — Python · FastAPI · ChromaDB · Claude API · Docker · Render **Jan 2026 – Present**

Live demo: rag-docs-assistant-e96h.onrender.com · *GitHub:* github.com/VivekGupta1999/rag-docs-assistant

- Built a retrieval-augmented assistant over FastAPI's official documentation (155 pages, ~2,400 chunks) with grounded answers, source citations, and refusal of out-of-scope questions.
- Built an evaluation harness from scratch: 30 test questions, retrieval hit@k, and two LLM-as-judge metrics — correctness against key facts and faithfulness against the retrieved context — using forced tool-use output and cached verdicts to control API cost.
- Ran 10 versioned experiments, one variable at a time: correctness 26% → 87%, faithfulness 35% → 83%, retrieval hit@5 74% → 96%.
- Improved ingestion: inlined code samples the docs reference from separate files, built heading-aware chunking that never splits a code block, labelled chunks "Page > Section", and used metadata filtering to keep changelog and API-reference pages out of retrieval.
- Built a tool-calling agent on the Claude API (search and page-read tools, capped loop): retrieval 23/23, but code stitched across pages lowered faithfulness, so v9 stays live.
- Tested hybrid search (BM25 + vectors) and cross-encoder reranking (4 models); both lowered quality and were not shipped. Traced failures to code held in adjacent chunks and added neighbour-chunk expansion: correctness 61% → 87%, faithfulness 65% → 83%.
- Containerized with Docker and deployed on Render with per-IP rate limiting; root-caused and fixed four production issues, including a data-store naming collision that caused an outage and a reverse-proxy IP-detection bug that broke rate limiting.

PROFESSIONAL EXPERIENCE

Backend Engineer — PPI Advisory **Feb 2022 – Dec 2025**

Toronto, ON, Canada · High-net-worth division of a national Canadian insurance marketing organization

Distributed Email Classification Service · C# · .NET · Microsoft Graph API · SQL Server · Azure

- Identified a critical performance bottleneck in the existing Outlook plugin and independently proposed, architected, and delivered a distributed, event-driven microservice, approved by stakeholders after a formal

system design review.

- Built a webhook-based pipeline: Microsoft Graph API subscriptions triggered sender lookups against SQL Server, revenue-based priority scoring, and classification tag application, entirely server-side. Implemented exponential backoff retry logic (up to 2 hours) for zero data loss during API throttling.
- Processed 60,000–80,000 emails/day across 600 employees, extending support to shared mailboxes unavailable in the previous architecture, and eliminated manual morning email triaging for helpline teams company-wide.
- Wrote unit tests covering core classification logic and edge cases; monitored system health via Azure Monitor to proactively catch anomalies and maintain high availability.

Commission Calculation Engine & Financial Reporting · SQL Server · C# · .NET · Power BI

- Owned a concurrent SQL Server stored procedure system for daily advisor commission calculations, with temp table caching, structured error/warning/info logging, and unit-tested release validation.
- Handled multi-company rate structures, secretary email attribution, and year-end tier-based payout classifications for hundreds of financial advisors nationwide, ensuring accurate financial reporting.
- Built a REST API consumed by the finance team's internal portal for real-time commission data, replacing manual spreadsheet reporting, and integrated results into organization-wide Power BI dashboards.

Backend Automation & Identity Systems · PowerShell · Azure · Active Directory · SQL Server

- Built a resumable data archival system migrating ~200GB of hybrid on-prem/Azure storage by file access age, with full directory structure preservation and audit logging, resumable from checkpoint if interrupted.
- Developed a distributed identity sync system linking 3 enterprise applications on a shared employee email key, handling concurrent updates and edge cases (e.g., post-marriage email changes), auto-pausing 100-150 subscriptions for employees on leave, and generating HR reconciliation reports.
- Built an Active Directory-integrated service account lifecycle system, automatically disabling inactive accounts with full audit trails to reduce licensing overhead.

Application Support, Observability & Delivery

- Monitored production services via Azure Monitor and IIS log analysis, proactively diagnosing root causes and resolving incidents before business impact.
- Managed Agile delivery using Jira, tracking sprint tasks and reporting progress in weekly stakeholder meetings across the full SDLC.

Software Developer Intern — Sheridan College **May 2021 – Sep 2021**

Oakville, ON, Canada

- Developed features for a mentor-mentee web application over a 4-month internship, applying OOP principles across the full SDLC including design, testing, and code review.

OTHER PROJECTS

- **SaleSpotter** — Node.js, Express, React, TypeScript, MongoDB Atlas, JWT: full-stack marketplace platform with 4 core REST modules (authentication, ad management, notifications, geospatial search). Implemented JWT-based authentication and role-based access control, plus MongoDB geospatial queries for location-based discovery.

CORE SKILLS

AI / LLM: RAG, Vector Search (ChromaDB), Hybrid Search, Reranking, Tool-Calling Agents, LLM Evaluation (LLM-as-judge, hit@k), Prompt Engineering, Claude API
Languages: Python, C#, JavaScript, TypeScript, SQL, PowerShell
Backend & Cloud: FastAPI, .NET / .NET Core, Node.js, Express.js, REST APIs, Microservices, Webhooks, Microsoft Graph API, JWT Auth, Docker, Azure (Monitor, DevOps, IIS), Render, Active Directory
Data: SQL Server (stored procedures, query optimization), MongoDB, ChromaDB, Data Pipelines, Power BI
Practices: System Design, Unit Testing, Code Review, Agile / Jira, CI/CD, Full SDLC

EDUCATION & CERTIFICATIONS

Honours Bachelor of Computer Science — Sheridan College, Oakville, ON · Dec 2021 · GPA 3.00/4.00
Azure Developer Associate (AZ-204) — In Progress · **Anthropic Academy** — Claude API, MCP, Claude Code